

Intervals and absolute value equations

Polynomial-size vertex test for inverse M-matrix intervals

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Rating rationale: Replacing an exponential interval test by a quadratic vertex family is challenging; verified inverse-positivity has specialist importance in interval matrix analysis.

Area: interval linear algebra; structured matrices

Independently reviewed resolution - 2026-09-11

Affirmative resolution. Theorem 1 proves that every interval member is inverse-M if and only if the n^2 vertices $C - D_i R D_j$ are inverse-M. These are contained in the displayed two-sign family, so the original $2n^2$ equivalence follows. The proof covers all real endpoints, every dimension, zero widths, zero entries and reducible matrices without assuming regularity.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the [complete manuscript](#), [independent agent review](#) and [submission record](#). The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. This is agent verification, not external human peer review or formal proof-assistant certification.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Lean verification — 15 September 2026

The complete original question has a formally verified affirmative answer: for every positive dimension and every ordered pair of real entrywise interval endpoints, the n^2 negative-sign vertices characterize the entire interval consisting of inverse M-matrices. This proves the full original two-sign criterion. Zero widths, zero entries and reducible matrices are included; no interval regularity or nonsingularity assumption is added. The manuscript's arithmetic and bit-complexity claims are outside this formalization.

All four reviewed statements passed LeanCert kernel trust audits, standard-axiom checks, and [fresh isolated Linux Comparator/default-kernel verification](#) at the [immutable proof revision](#). Two independent statement reviews preceded implementation and two independent nonauthor final code reviews passed. Actual rejection and sandbox controls passed.

Formalization: Sidney Holden, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship as recorded in the source. Reviews are AI-agent reviews, not external human peer review or official Tau Ceti endorsement; no source-author endorsement of the formalization is asserted.

[Proof and reviews](#) · [Retained Linux evidence](#).

Problem statement

A real square matrix M is a nonsingular M-matrix if its off-diagonal entries are nonpositive, M is invertible, and M^{-1} is entrywise nonnegative. An inverse M-matrix is the inverse of such a matrix.

For every $n \geq 1$, let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise, and set

$$\mathcal{A} = \{A : L \leq A \leq U\}, \quad C = (L + U)/2, \quad R = (U - L)/2.$$

For $i = 1, \dots, n$, let $z^{(i)}$ have entry -1 in position i and entry 1 elsewhere, and write $D_i = \text{diag}(z^{(i)})$.

Is the following equivalence true?

$$\begin{aligned} & [\text{every } A \in \mathcal{A} \text{ is an inverse M-matrix}] \\ & \iff \\ & \left[\begin{array}{l} C + sD_iRD_j \text{ is an inverse M-matrix} \\ \text{for every } i, j \in \{1, \dots, n\} \text{ and } s \in \{-1, 1\} \end{array} \right]. \end{aligned}$$

All entries in the interval vary independently, and degenerate intervals are allowed. The proposed test uses at most $2n^2$ matrices; repeated test matrices are harmless. This would give a small exact certificate for a structured property needed in interval inversion and verified linear-system computation.

References

Milan Hladík, *An overview of polynomially computable characteristics of special interval matrices*, in *Beyond Traditional Probabilistic Data Processing Techniques*, Springer (2020), pp. 295–310; [author preprint](#), [arXiv:1711.08732v1](#), §9, Conjecture 1, p. 11, with the vectors defined in Theorem 24.

Jürgen Garloff, Doaa Al-Saafin, and Mohammad Adm, *Further Matrix Classes Possessing the Interval Property*, *Reliable Computing* **28** (2021), 56–70, p. 64, paragraph immediately before IP 4.4.

Status check

The 2021 paper explicitly retains this conjecture while proving a different sufficient vertex family of size 2^{n-1} . On 2026-09-10, searches combining the original title, inverse M-matrices, interval property, conjecture, and 2025/2026 found no resolution. Hladík's current publication list and the later interval-property paper were checked. This is a selected-source status check, not an exhaustive citation audit.

Independent audit — 2026-09-10

The full Hladík preprint, §9, Conjecture 1, agrees with the displayed two-sign vertex family, including the definition of each $z^{(i)}$. Garloff–Al-Saafin–Adm, p. 64, expressly leaves that conjecture unresolved; its exponential family is a different test, not a resolution for a subclass here. Independent later searches for the exact conjecture and matrix class found no resolution.